ISSUE #01

Monthly Newsletter

CATALYST

DECEMBER 2025

EDITOR'S NOTE

Here's what has happened in the last three months and what's to come!

Welcome to our monthly team newsletter! We are thrilled to share with you the latest developments in our mission to make a positive impact in our community.

Firstly, we would like to express our gratitude to our Principal, Dr. P.B. Mane and our faculty coordinator Prof. Venkat Ghodke. Your generosity and dedication have allowed us to keep our tasks running smoothly and effectively.

In terms of our recent activities, we are nearing the end of the initial projects taken up as finding a solution to certain problems teachers faced in the institutions. Now the focus will shift on upcoming hackathons and the outreach program to get real industrial projects to Catalyst and get industry exposure.

We are also excited to announce that Catalyst will continue expanding its technical initiatives throughout the coming month. Our team is working on new research driven content, collaborations with other committees, performance deep dives, curated tech insights and student focused skill development. We look forward to bring high-quality projects and material that reflects Catalyst's commitment as the Think Tank of IOIT.

Finally, if you are interested in contributing ideas, collaborating with our teams and contributing to projects that actually matter, we welcome you to reach out.

Enjoy this month's newsletter!
Created by Yashada Rane

Catalyst Team x

In this newsletter you can expect:

- Hackathon Wins
- Documentation & Knowledge Management System (DKMS)
- Indigenous Function Generator
- India's First Fully Open Source Tri-Mode FCU in the Making

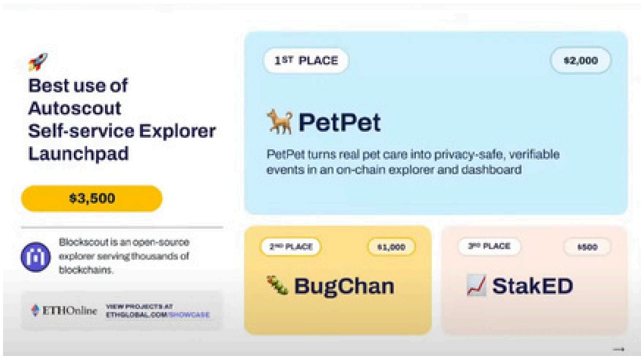
Arnav’s Solo Victory at Avalanche Pitch Day

Arnav Dandawate (CTO) represented Catalyst at the Avalanche Team1 Hackathon in Pune, where he secured a 1000 USD prize, competing solo in a field where most participants arrived with full teams.
His project, Glacier, applies a robust model where files are split into thousands of encrypted parts, distributed across independent provider machines, and reassembled securely using user-held keys. Glacier also incorporates an economic layer using AVAX, enabling storage payments, provider incentives, and staking mechanisms that ensure reliability.

His performance earned him not only a first-place victory but also an invitation to pitch at the Avalanche Pitch Day on November 3,2025 in Goa where he presented Glacier to venture capitalists and industry leaders. This achievement reflected both Arnav’s individual excellence and the growing technical capability within the Catalyst community, reinforcing the club’s commitment to building, executing, and delivering impactful engineering work.

Neural Nexus Secures Rank 2

Team Neural Nexus from Catalyst delivered an exceedingly well performance at Claude Solve-A-Thon 2025 at IIIT Nagpur on October 10-11 2025, securing Rank 2. The team consisted of Namit Solanki (CEO), Shaahid Shaikh (Tech Exec), Meghraj Jare and Harsh Surve and they were awarded 10,000 INR as the prize. Their project GreyMatter stood out amongst the competing entries for clarity of the problem statement and thoughtfully designed user centric approach.
This win marked the beginning of a series of wins for Catalyst, motivating others as well as reinforcing the message to the entire institution that Catalyst was created to Build, Deliver & Execute.



BugChan at ETHOnline 2025

Catalyst is pleased to highlight the success of Team BugChan, which secured 2nd place at EthOnline 2025, one of the most competitive international hackathons hosted by ETHGlobal. The project was awarded for the innovative application of the AutoScout Explorer, earning a prize of 1000 USDC. BugChan is a decentralized bug bounty platform that lets different organizations to start bug bounty programs and have bug reporters pay a stake amount before they create reports. The team consisted of two members from Catalyst: Manasi Choudhari (Vice-President) and Sarvesh Kolte (Tech Exec).

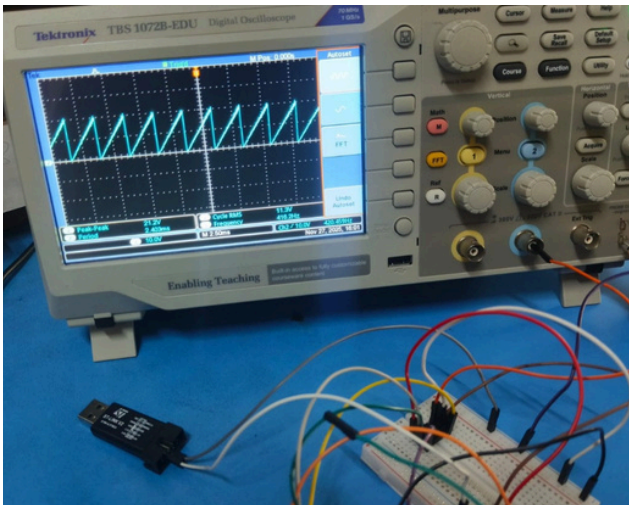
Documentation & Knowledge Management System (DKMS)

Catalyst is developing a Documentation & Knowledge Management System (DKMS) designed to streamline how college faculty and administrative teams manage documents. Documents stored across multiple platforms, files and folders are inefficient for the respective parties dealing with it. DKMS aims to centralize document storage for institutions while offering a clean and organized user friendly application for users to store, search and manage their files efficiently.

The system is built on TrueNas as the primary storage solution with MinIO providing compatible API points to create reliable storage and management system.

A Rust-based backend server sits between the client and the MinIO storage layer, handling authentication, permissions, metadata, and request validation. This backend exposes simplified REST APIs for the client, which is built using React. The client interacts only with the backend, providing users an intuitive interface for uploading, organizing, and managing documents.

The TrueNAS and MinIO setup has been completed, and ongoing work is focused on the backend services and frontend development. The project is being led by Sarvesh Kolte(Tech Exec) and once completed, DKMS will provide institutions with a robust, scalable, and user-focused solution for knowledge management.

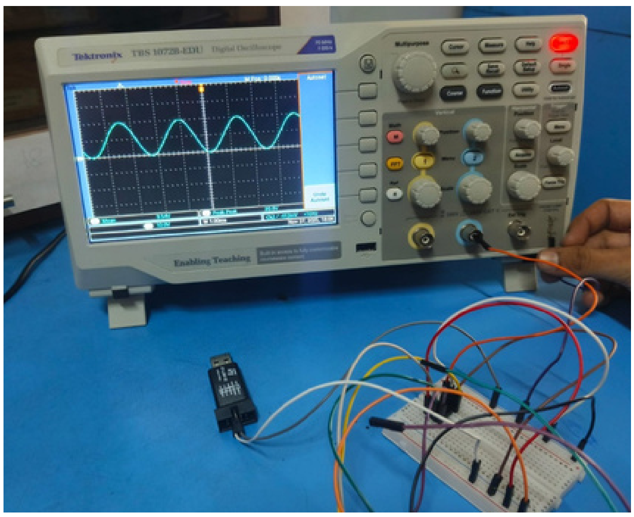


Indigenous Function Generator

The STM32-based function generator is being built focusing on accuracy, stability and low cost. The project began by finalizing the performance requirements, which included generating sine, square, and triangle waveforms from 1 Hz to 100 kHz with an adjustable 0–5 V peak-to-peak output. These fixed specifications guided the entire design. At the core of the system is the STM32F411 microcontroller, responsible for digitally synthesizing all waveforms. A 12-bit MCP4921 DAC converts digital samples into analog signals, which are then filtered through a precision RC reconstruction filter and amplified using a low-noise rail-to-rail operational amplifier.

Once assembled, the device underwent thorough testing using an oscilloscope to verify frequency accuracy, amplitude linearity, and waveform purity. After successful calibration, the entire system was housed in a compact aluminium enclosure, completing a robust and low-cost single-channel function generator suitable for educational and experimental use.

The project is currently 75% complete, with successful generation and verification of all four waveforms sine, square, triangle, and sawtooth on a digital storage oscilloscope directly from the STM32-based system. Remaining work focuses on enclosure design, final calibration, UI refinements, and extended feature validation. The project is being developed by Arya Nirhali(Tech Exec), Sanket Kolhe and Vedant Nalawade, and is on track to deliver a robust, low-cost function generator suitable for educational and experimental applications.



Catalysts x
Team AeroGuardians

India's First Fully Open Source Tri-Mode Flight Control Unit in the Making

Marut is India's First Fully Open Source Tri-Mode Flight Control Unit, designed to operate fixed wing aircraft, quadcopters and VTOL platforms with a single unified architecture. It is the outcome of a partnership between IoIT Drone Club's Team Aeroguardians and the Catalyst Committee, who share a common goal of making advanced aerial systems accessible to everyone. The project is being led by Aryan Basnet (President) accompanied by his team members Siddesh Kavitkar, Sharal Vishwakarma, Yash Tawar, Shreyas Karade, Karan Tikoo and Pushkar Lokhande.

Marut's basic premise is straightforward. Expensive proprietary hardware shouldn't be a barrier to high performance control systems. To achieve this goal, Marut is engineered around the extremely affordable yet highly capable STM32F411CEU microcontroller. This chip becomes remarkably powerful when paired with a carefully optimized FreeRTOS environment that unlocks fast task scheduling, reliable sensor handling and a level of safety normally seen only in far more expensive flight controllers.

Marut prioritizes cost without compromising functionality. In the Indian market, there isn't currently a consumer flight controller that provides support for all three major platform types fixed wings, quads and VTOL aircraft within a budget of only 3000-4000 INR.

Marut is designed to empower student teams and researchers by providing professional flight control capabilities at a fraction of the typical cost. By removing financial barriers, we enable developers to explore aeromodelling and autopilot innovation without being constrained by expensive proprietary gear.

The Marut Flight Control Unit (FCU) has reached a key stage of development with a stable multi-threaded RTOS architecture now running core functions such as sensor fusion, control loops, RC input, telemetry and mode switching. All major peripherals are operational, including timers for PWM and PPM capture, UART ports for GPS and telemetry, I2C for the IMU, magnetometer and barometer and ADC for battery monitoring. Both quadrotor and fixed-wing modes are active through dedicated threads, and the quad PID controller is functioning with calibrated gyro data, motor mixing, safety resets and reliable servo and ESC outputs.

GPS reception and NMEA parsing are operational, and the telemetry system is adept at sending MAVLink-style messages with active LED and buzzer indicators. The project stands out for achieving high performance on a minimal processing platform, demonstrating that a well-architected system with precise task management can enable a low-cost microcontroller to effectively support an evolving autonomous system.

Thank you for reading!